

# The race between hydrogen and heat pumps for space and water heating: A model-based scenario analysis

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## ABSTRACT

This paper analyses different levels and means of the electrification of space and hot water heating using an explorative modelling approach. The analysis provides guidance to the ongoing discussion on favourable pathways for heating buildings and the role of secondary energy carriers such as hydrogen or synthetic fuels. In total, 12 different scenarios were modelled with decarbonisation pathways until 2050, which cover all 27 member states of the European Union. Two highly detailed optimisation models were combined to cover the building stock and the upstream energy supply sector. The analysis shows that decarbonisation pathways for space and water heating based on large shares of heat pumps have at least 11% lower system costs in 2050 than pathways with large shares of hydrogen or synthetic fuels. This translates into system cost savings of around €70 bn. Heat pumps are cost-efficient in decentralised systems and in centralised district heating systems. Hence, heat pumps should be the favoured option to achieve a cost-optimal solution for heating buildings. Accordingly, the paper makes a novel and significant contribution to understanding suitable and cost-efficient decarbonisation pathways for space and hot water heating via electrification. The results of the paper can provide robust guidance for policymakers.

## 1. Introduction

The heating and cooling sector accounts for about 50% of European total final energy consumption, significantly more than the electricity and transport sectors [1]. At the same time, the heating and cooling sector is still largely based on fossil fuels in most European countries [2]. Thus, rapid progress in decarbonising the heating and cooling sector has the potential to make a significant contribution to achieving the European Union's (EU) target of climate neutrality in 2050 [3]. Given the current crisis, decarbonising the heating sector is even more important to reduce the dependence on natural gas imports from Russia.

The feasibility of reaching a climate-neutral energy system has been

discussed extensively in the literature. Several options are provided, usually as the result of modelling different scenarios or transformation pathways (see overviews in [4–6]). In the existing literature, there seems to be general agreement on the pathway for the power sector with the phase-out of fossil fuels and rapid expansion of renewable electricity, mainly from wind and solar [4,6]. Regarding heating and cooling, however, there is less consensus and different decarbonisation pathways have been suggested, the impacts of which are still being discussed. For space and hot water heating, electrification is seen as one of the main solutions by several studies [4,5,7–10], also expressed in the concept of smart energy systems [11–13]. Overall, there is a strong tendency towards a consensus in the scientific literature that heat pumps are more

**Abbreviations:** CHP, Combined heat and power; COP, Coefficient of performance; CSP, Concentrated solar power; EU, European Union; MENA, Middle East and North Africa; MILP, Mixed integer linear programming; O&M, Operation and Maintenance.

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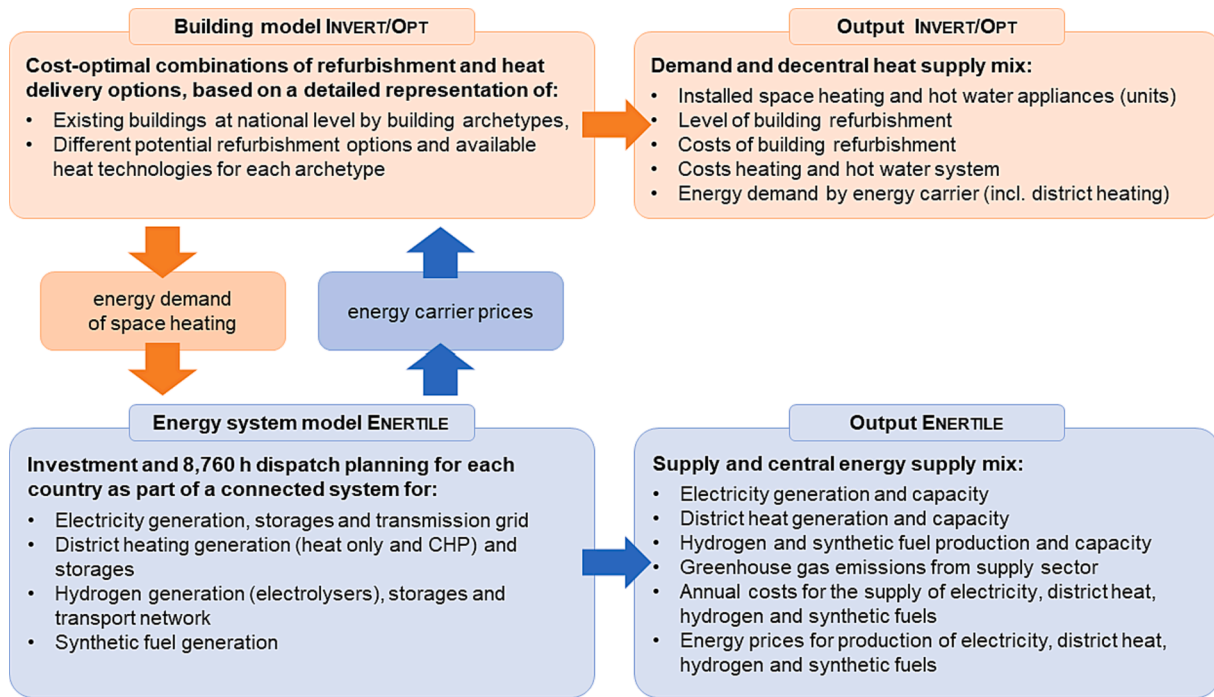


Fig. 1. Overview of the methodology with the combination of the two models INVERT/OPT and ENERTILE.

efficient and economical than hydrogen and synthetic fuels for space and water heating (compare review [14] and e.g. [15–19]). However, there is still a heated political debate about the role of hydrogen and synthetic fuels and there are a few studies that foresee a greater use of hydrogen or e-fuels (see e.g. [20–22]). In particular, some players such as gas distribution operators or gas boiler providers see a high contribution of hydrogen in space and water heating [23,24]. To date, there are only very few project reports that underpin the role of hydrogen in space and water heating in an integrated European energy system by providing concrete numbers (see, e.g. [8,17,25]). Due to the heterogeneity of buildings, assessing the impact of different pathways to a carbon neutral heating sector requires highly detailed modelling of the building sector. At the same time, a comprehensive energy system model needs to be applied to consider the impact of heating choices on the whole energy system. To the best of our knowledge, this paper is the first scientific publication to analyse the technology-specific effects of changes in space and water heating as well as the impact on the upstream supply sector for the whole EU.

This paper contributes to the literature by providing an in-depth and detailed model-based analysis comparing the total system costs of different levels of direct electrification based on heat pumps or electric boilers and indirect electrification based on low carbon secondary energy carriers such as hydrogen or synthetic gases or liquids. The main research question of this paper is: What is the most favourable electrification pathway for heating buildings, taking into account the impact on the upstream supply sector?

The analysis uses an explorative scenario approach to assess a wide range of pathways and to provide a complete picture of the different options still under discussion. In total, 12 different scenarios are modelled with pathways until 2050, covering all 27 member states of the EU. To address the specific challenges of modelling both the building sector in detail and the energy system as a whole, two models have been coupled in this assessment. The mixed-integer optimisation model INVERT/OPT is based on a detailed representation of the building sector for the whole EU and finds cost-optimal combinations of renovation and heat supply options. It is combined with the linear optimisation model ENERTILE, which has a detailed geographical coverage of renewable energy potentials and optimises the supply of electricity, district heating

and synthetic fuels and gases for the whole EU. With this highly detailed modelling approach, the paper makes a novel and significant contribution to understanding long-term perspectives and optimal technology mixes for space and water heating.

The remainder of the paper is structured as follows: Section 2 describes the overall modelling approach and the models used, defines the scenarios and describes the main data sources and assumptions. Section 3 presents the main results, i.e. the modelling output at EU-27 level for decentralised heating, district heating, electricity generation and imports as well as system costs and investments required. Section 4 discusses key results in the context of existing literature. Section 5 provides conclusions and policy implications.

## 2. Methodology

### 2.1. General framework and modelling approach

In this paper, an explorative scenario-based modelling approach, coupling the two highly detailed models INVERT/OPT and ENERTILE, is used to answer the research question outlined above. The chosen explorative approach serves this objective with a high number of different scenarios and thereby scans a wide solution space.

In scenario-based analyses, multiple scenarios are developed, each based on a different set of underlying assumptions, to investigate how energy systems react to these changes. By comparing and contrasting these scenarios, it becomes possible to evaluate the factors influencing the diverse outcomes and identify the pathways that are more advantageous for the energy system in terms of costs and investments. This approach allows for a comprehensive exploration of potential future trajectories and facilitates informed decision-making regarding the most optimal and economically viable pathways for the energy sector. Hence, in our analysis, this approach allows us to compare a wide range of heating options and evaluate their impacts on energy system costs and configurations.

The two models are used in an iterative order: First, INVERT/OPT generates the demand datasets and a preliminary decentralised heat supply mix, then ENERTILE produces the centralised energy supply and energy system datasets. For consistency, the energy prices derived from

ENERTILE are then returned to INVERT/OPT through an iterative feedback loop impacting the final decentralised heat supply mix. For all 27 member states of the EU, the pathway from 2030 until 2050 is modelled in 10-year steps. Fig. 1 presents an overview of the methodology and the combination of the two models. In addition, Figure A 1 in the appendix presents a methodological overview of the building model INVERT/OPT and Figure A 2 in the appendix visualises the framework and methodology of the energy system model ENERTILE. In the following, the two models are described in detail.

## 2.2. Building model INVERT/OPT

INVERT/OPT is based on a detailed representation of existing buildings at national level by building archetypes, as well as different potential refurbishment options and available heat technologies for each archetype.<sup>1</sup> A mixed integer linear programming (MILP) algorithm is applied to find cost-optimal combinations of refurbishment and heat delivery options under given constraints.<sup>2</sup> For each country, the optimisation function identifies the combination of refurbishment option  $k$  and supply system change  $j$  in each building segment  $i$  that leads to the lowest overall system costs at a given time in the future starting from the currently existing status (compare formula 1). Further and more detailed descriptions of the INVERT/OPT model can be found in [27].

$$\min \sum_i \sum_{j,k} (cost_{supply,j,i} + cost_{refurb,k,i}) \quad (1)$$

with

$$cost_{supply,j,i} = cost_{invest,j,i} \cdot CRF_j + cost_{O\&M,j,i} + cost_{energy,j,i}$$

$$cost_{refurb,k,i} = cost_{invest,k,i} \cdot CRF_k$$

and

$cost_{invest,j,i}$ : Investment costs for supply system change  $j$  in building segment  $i$

$cost_{O\&M,j,i}$ : Operation and maintenance costs for supply system change  $j$  in building segment  $i$

$cost_{energy,j,i}$ : Energy costs for supply system change  $j$  in building segment  $i$

$CRF_j$ : Capital recovery factor for supply system change  $j$

$cost_{invest,k,i}$ : Investment costs for refurbishment option  $k$  in building segment  $i$

$CRF_k$ : Capital recovery factor for refurbishment option  $k$

The building stock in the model is defined based on three input databases for each country: a database of the existing building stock, a database of heating and hot water supply technologies and a database of refurbishment options. In addition, country-specific energy prices,

the INVERT model family has been compiled over the last decade over the course of numerous projects. The Tabula/Episcopo building stock database for residential buildings is an important data source [31]. A representation of the data contained in the INVERT database on existing building stocks can be found in [32–34].

The data from the databases are used together with climate data (average monthly temperatures, sunshine) and user behaviour (time of use, set indoor temperatures) to calculate energy needs and delivered energy for space heating and hot water production. The resulting energy consumption, cost and emission parameters are calculated for all possible combinations of delivery systems and renovation options in the different building segments in the countries analysed. About 500 to 1000 options are considered for each building segment. The proportion of buildings in each archetype that implement a renovation option between the baseline and target years of the analysis is determined endogenously by the model. It is based on the distributions of the construction or past renovations and the Weibull distribution for the lifetime of the components. The computed properties of all possible options in all building archetypes, together with constraints on the model solution allowed form the input data for the upcoming optimisation part of the model. For each country, the INVERT/OPT optimisation module identifies the combination of upgrade and power system change options in each building segment that yields the lowest total system cost at a given point in the future, starting from the current state.

The following constraints are defined in the INVERT/OPT model: (a) lower and upper share of floor area that applies the maintenance option (as opposed to the thermal renovation option) defined on the country level; (b) lower and upper share of floor area that applies the maintenance option defined on the level of individual building cohorts; (c) upper limit of CO<sub>2</sub> emissions, (d) upper and lower share of heated floor area per energy carrier group such as gaseous energy carriers, liquids, electricity, biomass, district heating; (e) upper limit for the energy carrier-specific energy consumption.

## 2.3. Energy system model ENERTILE

ENERTILE is a detailed techno-economic linear optimisation model for energy systems, which minimises the costs of energy generation, transmission and storage [35]. The objective function minimises all costs associated with the vectors of the decision variables for installed capacity  $\vec{X}$  and generation  $\vec{x}$ . For reasons of space and clearness, only an excerpt and simplified version of the objective function of the linear optimization model is presented (compare formula 2). Detailed descriptions of ENERTILE can be found in [36–38] and all relevant formulas of the optimisation problem can be found in [39].

$$\min_{\vec{X}, \vec{x}} \left[ cost_{el}^{fix}(\vec{X}) + cost_{el}^{var}(\vec{x}) + cost_{heat}^{fix}(\vec{X}) + cost_{heat}^{var}(\vec{x}) + cost_{el,chp}^{var}(\vec{x}) + cost_{H2}^{fix}(\vec{X}) + cost_{H2}^{var}(\vec{x}) + cost_{syn-fuels}^{fix}(\vec{X}) + cost_{syn-fuels}^{var}(\vec{x}) \right] \quad (2)$$

availability constraints and resource restrictions for energy carriers are specified.

The database on existing building stocks that is the common basis for

with

$cost_{el}^{fix}$ : Fixed costs of electricity capacity expansion in €

$cost_{el}^{var}$ : Variable costs of electricity generation in €

$cost_{heat}^{fix}$ : Fixed costs of capacity expansion in heat grids in €

$cost_{heat}^{var}$ : Variable costs of heat generation in heat grids in €

$cost_{el,chp}^{var}$ : Variable costs of electricity generation from CHP plants in

heating grids in €

$cost_{H2}^{fix}$ : Fixed costs of hydrogen capacity expansion in €

$cost_{H2}^{var}$ : Variable costs of hydrogen generation in €

$cost_{syn-fuels}^{fix}$ : Fixed costs of synthetic fuel capacity expansion in €

$cost_{syn-fuels}^{var}$ : Variable costs of synthetic fuel generation in €

<sup>1</sup> The model description is based on [27].

<sup>2</sup> INVERT/OPT was developed based on the model INVERT/EE-LAB, which applies the same building stock archetypes, but uses a discrete choice logit-approach as a modelling logic (e.g. see [28–30]). The energy needs and energy consumption module, the module that calculates replacement rates based on lifetime distributions as well as the module that calculates the availability of replacement options along with the energy performance and cost implications are common to all INVERT model variants.

**Table 1**  
Overview of the modelled scenarios.

| Scenario group | Scenario name    | Energy carrier | Share of heated floor area supplied by energy carrier in 2050 |
|----------------|------------------|----------------|---------------------------------------------------------------|
| Elec           | <i>Elec 30</i>   | Electricity    | 30%                                                           |
|                | <i>Elec 40</i>   |                | 40%                                                           |
|                | <i>Elec 60</i>   |                | 60%                                                           |
|                | <i>Elec 80</i>   |                | 80%                                                           |
| H2             | <i>H2 20</i>     | Hydrogen       | 20%                                                           |
|                | <i>H2 40</i>     |                | 40%                                                           |
|                | <i>H2 60</i>     |                | 60%                                                           |
|                | <i>H2 80</i>     |                | 80%                                                           |
| E-Fuel         | <i>E-Fuel 20</i> | E-fuels        | 20%                                                           |
|                | <i>E-Fuel 40</i> |                | 40%                                                           |
|                | <i>E-Fuel 60</i> |                | 60%                                                           |
|                | <i>E-Fuel 80</i> |                | 80%                                                           |

ENERTILE covers the supply of electricity, district heating, and synthetic fuels and gases, i.e. hydrogen and methane. The model simultaneously optimises capacity expansion and hourly generation and dispatch of all system components based on the exogenously specified demand for electricity, district heating and synthetic fuels. The infrastructures modelled include conventional and renewable power plants, combined heat and power (CHP) plants, conventional and renewable heating technologies in district heating networks, cross-border electricity transmission capacities, energy storage technologies, hydrogen and methane generation technologies and infrastructures. Furthermore, the model considers demand-side flexibility from decentralised heat pump systems in buildings and electric mobility in the transport sector.

ENERTILE projects development paths for multiple years, which are optimised in a single model run using perfect foresight. At the same time, ENERTILE has a high temporal resolution with 8,760 h in each year analysed. ENERTILE geographical coverage for this paper comprises the 27 member states of the EU and each country corresponds to one model region, which is in turn part of the broader connected EU system. Infrastructures (electricity and gas grids) connect neighbouring countries. Covering a large region instead of single countries becomes increasingly necessary with high shares of renewable energy, as exchanging electricity between different weather regions is a central flexibility option [35]. The key constraints of the model ensure that the demand for electricity, heat and synthetic gases is met in every region and hour.

The heat supply in district heating networks and the associated capacity expansion of generation technologies is modelled in ENERTILE, while the heat demand is given exogenously for each region and year. In this paper, the annual heat demand is provided by the building model INVERT (see section 2.2).

ENERTILE scales the annual demand down to hourly demand, using daily district heating time series that incorporate average daily outdoor temperature. To meet the heat demand, different generation options are available including natural gas boilers, CHP plants, electric boilers, large-scale heat pumps, renewable heating technologies such as geothermal energy and solar thermal plants as well as biomass boilers and CHP, and heat storage. All large-scale heat pumps in ENERTILE are modelled as air-based heat pumps using ambient air. This approach can be justified by two facts: One, unlike other heat sources such as water or waste heat from industry, air-based heat pumps are independent of location and local conditions, and two, this is currently the predominant type of installed heat pumps [40]. This thus represents a conservative assumption, since in reality, more suitable heat sources with higher temperature levels and therefore higher efficiencies should be usable, at least to some extent. The efficiency of an air heat pump is strongly dependent on the outside temperature. Therefore, the coefficient of performance (COP) in the model is determined with a linear approximation function and the ambient temperature in hourly resolution [37]. The decisions about investments in and the dispatch of heating

technologies are part of the overall system optimisation.

The production of electricity-based fuels like hydrogen and e-fuels is optimised endogenously. Investments in electrolyzers as well as additional electricity production are part of the optimisation problem (see also [36]). For the supply of hydrogen, electrolyser technologies, hydrogen storages and hydrogen transport pipelines between countries are included. Generation technologies for e-fuels supply are also covered. Hydrogen and e-fuels can either be used to cover the demand for heating in buildings and other sectors, or they can be used in reconversion technologies such as hydrogen turbines and CHP technologies to produce climate-neutral electricity in times of shortages. Furthermore, these fuels can be used to supply heat in district heating networks. There is explicit coverage of interactions between the supplies of the different energy forms. Apart from the direct production of hydrogen and e-fuels using resources within the EU, it is also possible to import these fuels from outside the EU, e.g. from the Middle East and North Africa (MENA).

#### 2.4. Definition of scenarios

A set of 12 scenarios is defined, determined by the share of the heated floor area of the building supplied by the respective energy carrier, i.e. electricity, hydrogen and synthetic fuels. The 12 scenarios are divided into three different scenario groups: (1) direct electrification of space and hot water heating; (2) indirect electrification using hydrogen produced with renewable electricity; and (3) indirect electrification with synthetic e-fuels, which are hydrocarbon-based gaseous or liquid fuels produced using renewable electricity. Within each scenario group, four different scenarios are defined with different minimum shares of the specific energy carrier (see Table 1).

In each scenario, the target share is implemented as a model constraint in the building model INVERT/OPT at member state level. For example, in the *Elec 30* scenario, 30% of the heated floor area in 2050 in all EU member states must be provided by the energy carrier electricity and thus by heat pumps and electric boilers. The mix of technologies supplying the remaining heated floor area is based on least-cost optimisation. Thereby, district heating is restricted to areas with high heat densities, resulting in acceptable heat distribution costs.

All of the modelled scenarios achieve climate neutrality in 2050, not only in the space heating sector but in the overall energy system as well. Thus, a 100% climate-neutral energy system with high shares of renewable electricity and heat is achieved. All renewable energy sources are assumed to have zero carbon intensity. Thus, we implicitly assume that the remaining small shares of carbon-based energy supply (i.e. waste-to-energy) can be absorbed by natural carbon sinks.

All other assumptions remain constant across scenarios (e.g. assumptions for transport or industry). This means that technology-specific effects caused by changes in space and hot water heating can be identified, quantified and interpreted. The assumptions ensure that all other sectors achieve climate neutrality in 2050, e.g. with high shares of electric vehicles and decarbonisation of industrial processes (in line with the previous scenario analyses, compare Table A 8 in the Annex).

#### 2.5. Data and assumptions

The modelling work in both models is based on harmonised general assumptions. A uniform interest rate of 2% is assumed for all technologies in all scenarios. In addition, all scenarios are based on a sharply increasing CO<sub>2</sub> price of 65 €/t in 2030, 200 €/t in 2040 and 500 €/t in 2050 in order to achieve a carbon-neutral energy system in 2050. The fossil fuel prices for electricity and district heating supply used in all scenarios are presented in Table A 2 in the appendix. Thereby, the assumed prices for fossil fuels do not reflect the recent price increases, because the modelling work in this paper was completed before the start of the current energy crisis. The impact of higher prices is, however, discussed in section 4. Assumed techno-economic parameters for the

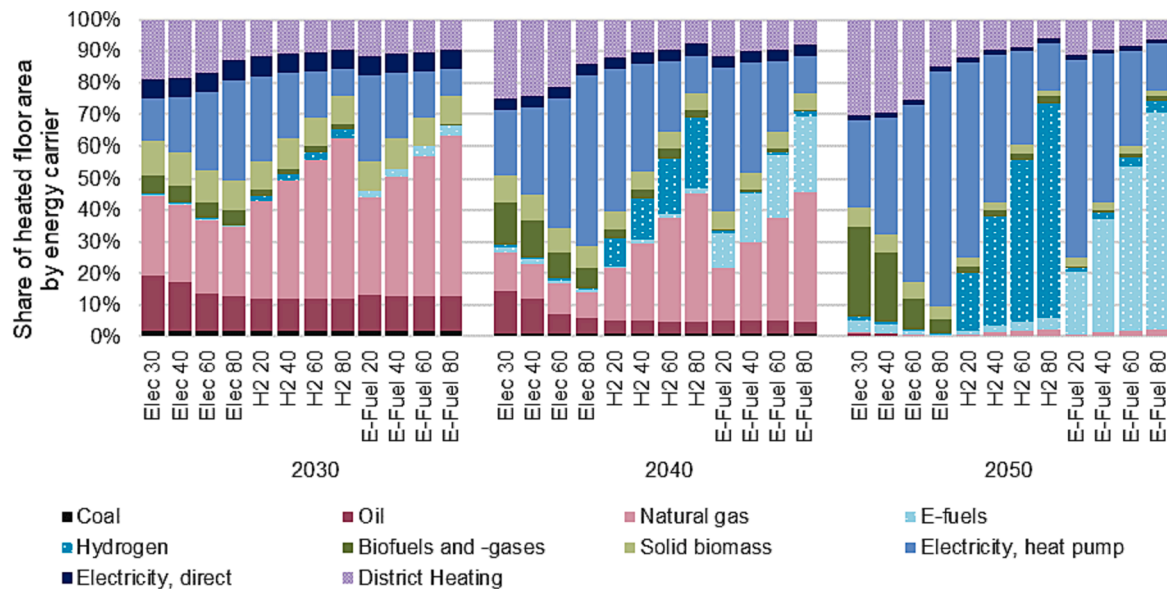


Fig. 2. Share of heated floor area by energy carrier in EU-27.

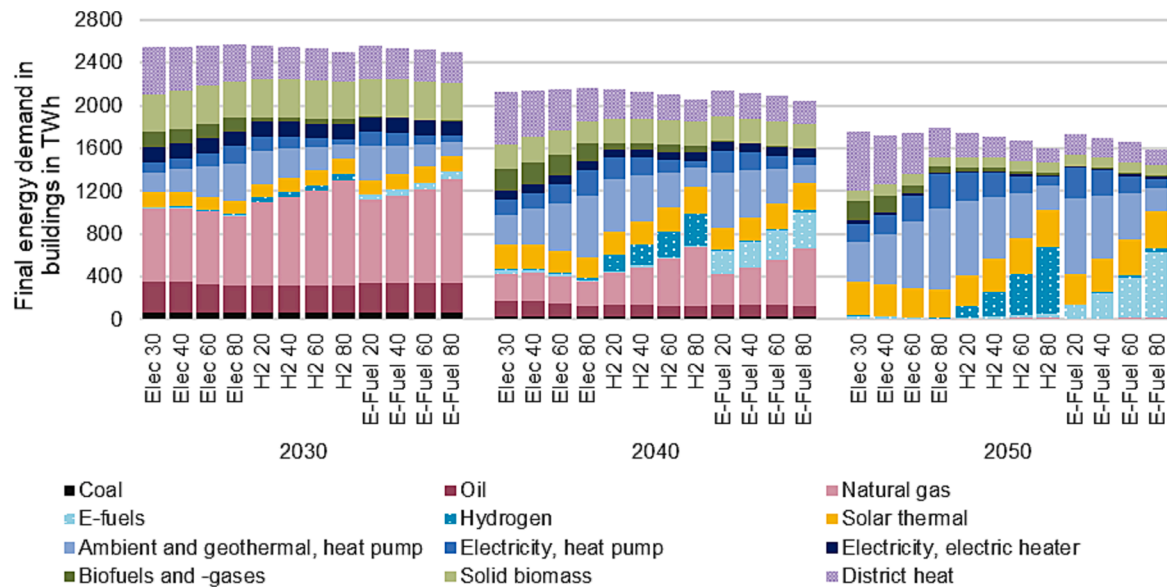


Fig. 3. Final energy demand for space and water heating by energy carrier in EU-27.

applied technologies, including assumptions on their future developments are presented in the appendix.

Hydrogen and synthetic gases are assumed to blend into the natural gas system and eventually replace it. The exact blending rate depends on the scenario. Losses incurred within energy distribution grids are also accounted for (5.5% for electricity, and 10% for district heating). Table A 8 and Table A 9 in the appendix present the input data for the demands of other sectors.

The modelling uses several restrictions for energy carriers. For biomass, country-specific upper limits are defined in decentralised and centralised heating (in line with [8]). In district heating, the limits for 2030 are in line with the current use of biomass [41]. By 2040, the limits are reduced to 75% and, in 2050, the limit is set to only 50% of the

current use of biomass. Municipal waste for electricity and district heating is set to the current use in 2030, limited to 90% in 2040 and 80% in 2050. Furthermore, for geothermal heat, it was assumed that countries currently using this technology will carry on deploying it [41]. In all scenarios, geothermal production was assumed to increase strongly (in line with [42]) and could reach 10 times today's production in 2050, as current production volumes are still very low. In the case of solar thermal energy, it was assumed that the technology is deployed in all countries, covering up to 10% of country-specific district heating demand in 2050, as almost all European countries have the potential for an economically viable use of solar thermal in district heating [43,44]. As solar and geothermal energy are not the focus of this analysis, their shares are constant in all scenarios. Finally, for renewable electricity

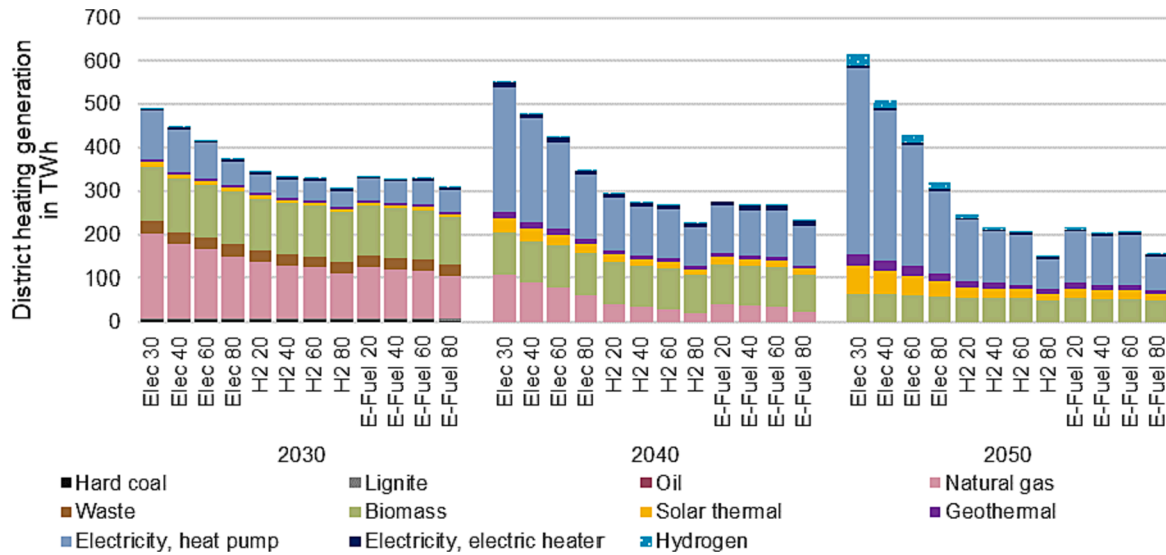


Fig. 4. District heating generation mix in EU-27.

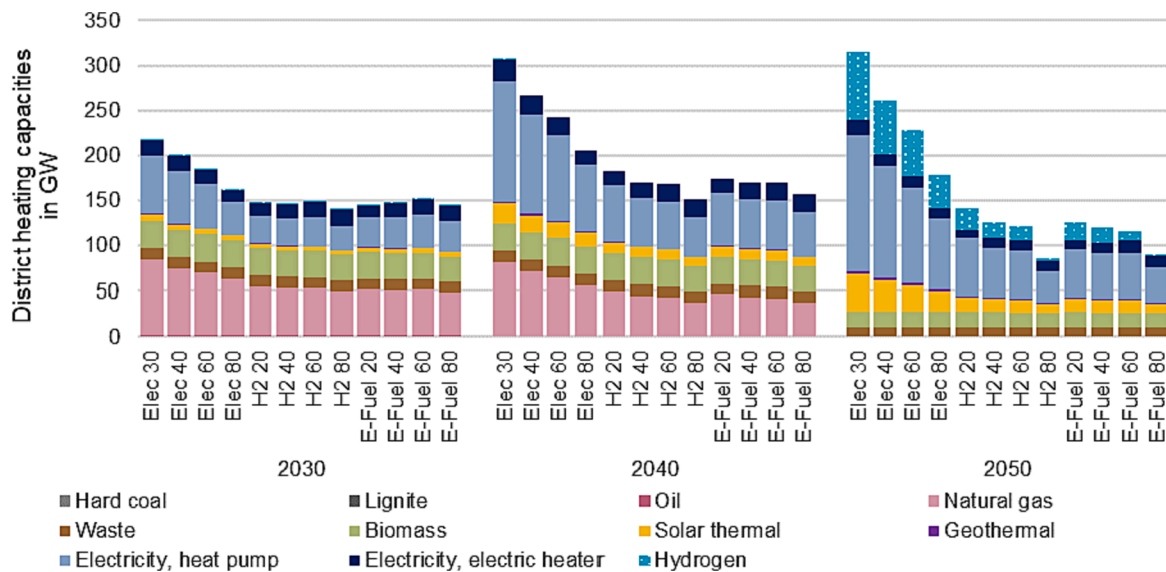


Fig. 5. District heating capacity mix in EU-27.

generation, lower limits were defined based on the current electricity generation in the countries [41]. In addition, upper limits for wind and solar were defined based on the generation potential calculated (see methodology in [35,37]). For electricity generation from hydropower and geothermal energy, it was assumed that their current production would remain constant until 2050 [41].

Under these assumptions and with the demand data, ENERTILE delivers hourly marginal costs for electricity, district heat, hydrogen, and e-fuel supply. These energy prices derived by ENERTILE are returned to INVERT/OPT through an iterative feedback loop, which started with an initial best guess based on previous model iterations and studies (e.g. [8]). For the calculation in INVERT/OPT, the wholesale prices delivered from ENERTILE were converted into retail prices, i.e. including taxes, grid fees, distribution costs and retail mark-ups.

### 3. Results

#### 3.1. Heated floor area and final energy demand in buildings

Fig. 2 presents the share of heated floor area by energy carrier and scenario in the EU-27.<sup>3</sup> It shows that in the *Elec* scenarios in 2050, especially in the *Elec 30*, *Elec 40* and *Elec 60* scenarios, the remainder of the heated floor area is covered by a high proportion of district heating. In the *H2* and *E-Fuel* scenarios, the remaining heated floor area is dominated by heat pumps, but district heating also contributes a substantial share. In all scenarios, biomass heating is reduced due to the assumed resource constraints (compare section 2.5).

It can be concluded from Fig. 2 that there is a clear hierarchy of

<sup>3</sup> The scenarios *H2 80* and *E-Fuel 80* show a slight underachievement of the targets for the heated floor area in 2050. This is due to different model restrictions, such as the expansion of hydrogen and gas grids which are in conflict with the minimum share of heated floor area covered by one energy carrier, partly triggered by limits on the replacement rates of existing heating systems.

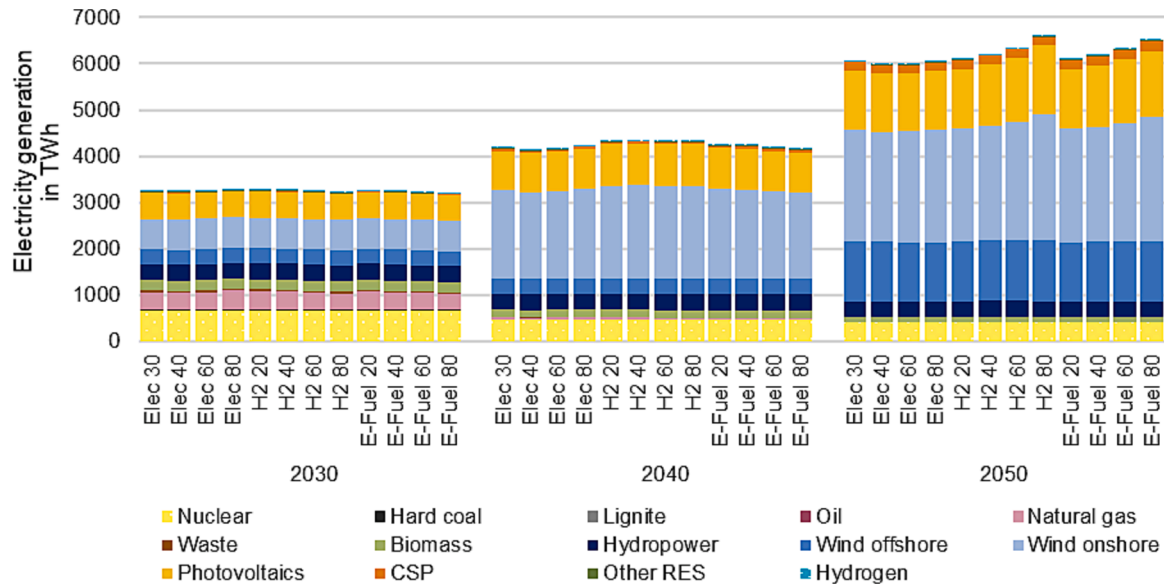


Fig. 6. Electricity generation mix in EU-27.

Table 2

Hydrogen and synthetic methane generation in EU-27 and imports from MENA region in 2050.

| Scenario name    | Hydrogen               |                   | Synthetic methane      |                   |
|------------------|------------------------|-------------------|------------------------|-------------------|
|                  | Generation EU-27 [TWh] | Import MENA [TWh] | Generation EU-27 [TWh] | Import MENA [TWh] |
| <i>Elec 30</i>   | 625                    | 0.000             | 551                    | 160 (22%)         |
| <i>Elec 40</i>   | 609                    | 0.000             | 552                    | 157 (22%)         |
| <i>Elec 60</i>   | 614                    | 0.000             | 542                    | 157 (22%)         |
| <i>Elec 80</i>   | 626                    | 0.000             | 537                    | 159 (23%)         |
| <i>H2 20</i>     | 714                    | 0.000             | 552                    | 147 (21%)         |
| <i>H2 40</i>     | 816                    | 0.001             | 565                    | 140 (20%)         |
| <i>H2 60</i>     | 926                    | 0.001             | 586                    | 127 (18%)         |
| <i>H2 80</i>     | 1130                   | 0.001             | 615                    | 111 (15%)         |
| <i>E-Fuel 20</i> | 607                    | 0.000             | 629                    | 190 (23%)         |
| <i>E-Fuel 40</i> | 601                    | 0.001             | 703                    | 219 (24%)         |
| <i>E-Fuel 60</i> | 598                    | 0.001             | 810                    | 261 (24%)         |
| <i>E-Fuel 80</i> | 603                    | 0.001             | 971                    | 325 (25%)         |

Note: Value in () shows the share of imports in the total supply.

heating technologies: Decentralised heat pumps and district heating are economically most viable, while hydrogen and e-fuels are relatively more expensive. This is due to the low efficiency of hydrogen and e-fuel production [45]. In other words, electricity-based synthetic fuels are an expensive form of energy due to the high conversion losses during production. While the retail energy prices of hydrogen and electricity are in a similar range, the electricity demand of heat pumps is comparatively low due to their high performance<sup>4</sup>. Consequently, the share of heat pumps is always quite high, even in the *H2* and *E-Fuel* scenarios.

Fig. 3 shows the final energy demand for space and hot water heating and the resulting energy carrier mix in the EU-27. It is apparent that all scenarios show a significant decrease in final energy demand until 2050.

<sup>4</sup> In the scenarios, we assume that buildings do not change their heat radiators. However, if a thermal renovation is carried out on the building, the model endogenously considers that the heat distribution temperature can be lower compared to the initial state. The temperature reduction that is possible due to the lower heat demand is determined based on a simplified logarithmic excess temperature concept.

This is reasoned by a general uptake in building renovation and the related energy savings. There are, nevertheless, small differences in the uptake of renovations. Renovation rates are somewhat higher in the *H2* and *E-Fuel* scenarios compared to the *Elec* scenarios. Hydrogen and e-fuels have the highest variable energy prices, which trigger a high level of renovation. However, heat pumps also show their highest COP in highly insulated buildings. Thus, ambitious levels of building renovation form part of the optimum.

### 3.2. District heating generation and capacities

Fig. 4 shows the heat generation and Fig. 5 shows the heat capacities in the district heating networks in the EU-27. In line with the INVERT/OPT results (see section 3.1), there is a higher amount of district heating in the *Elec* scenarios than in the *H2* and *E-Fuel* scenarios.

The generation mix in Fig. 4 shows that fossil fuels are phased out and that in 2050, heat pumps provide the majority of the heat. Even if the total district heating demand increases greatly, as in the *Elec 30* scenario, heat pumps can cover any additional demand, highlighting their relevance for future district heating systems. In 2050, heat pump capacities reach up to 150 GW in the *Elec 30* scenario. The current level of heat pumps in district heating is only 2170 MW [46], which means that the capacity of heat pumps needs to increase by up to 148 GW in the *Elec 30* scenario.

Complementary heat generation technologies in 2050 are biomass, solar thermal and geothermal energy plants. In all scenarios, biomass is reduced due to the assumed resource constraints (see section 2.5). Similarly, geothermal and solar thermal enter the mix in line with the assumptions (see section 2.5). Further, electric heaters and waste CHP plants only have small shares in the generation and capacity mix. Hydrogen-based heat generation technologies, primarily hydrogen boilers, reach quite high capacities, even though the generation is low. This indicates the backup role of hydrogen for district heating in times of electricity shortages. Overall, hydrogen is used less in the *H2* and *E-Fuel* scenarios compared to the *Elec* scenarios. In the *H2* and *E-Fuel* scenarios, synthetic fuels are required in decentralised heating applications. These have higher prices due to the high demand, and are consequently less available for cost-efficient use in district heating.

### 3.3. Electricity generation and imports

Fig. 6 illustrates the impact of different approaches for electrified

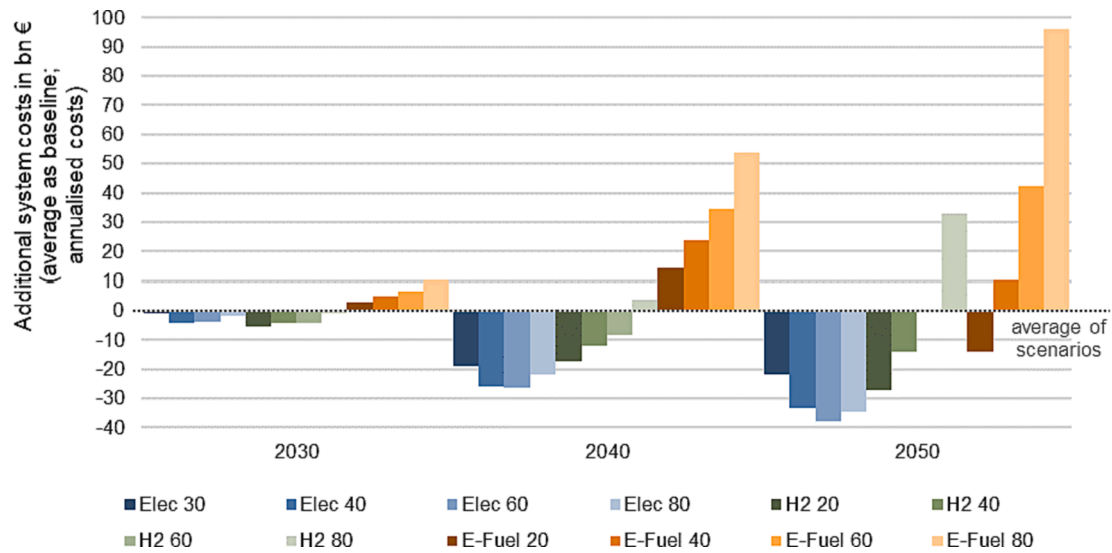


Fig. 7. System costs in EU-27 with average (per decade) of scenarios as a baseline.

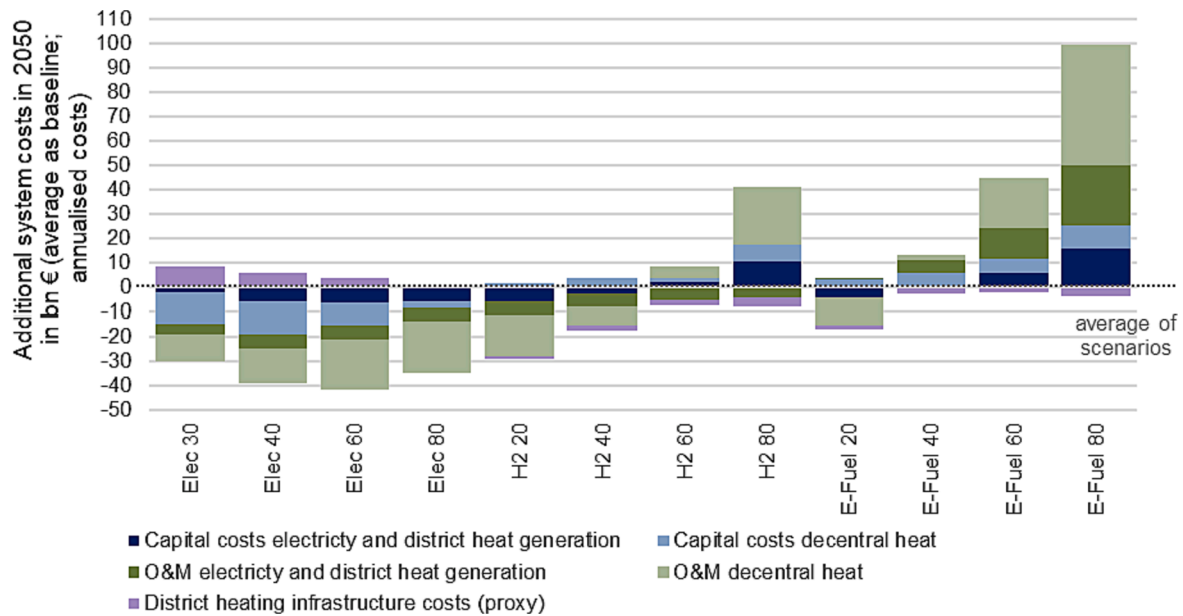


Fig. 8. System costs in 2050 in EU-27 divided into cost components with the average (per decade) of the scenarios as a baseline.

heating on the electricity generation mix in the EU-27. In all scenarios a phase out of fossil fuels and a significant increase in electricity generation can be observed until 2050. Even though the lowest overall generation occurs in the *Elec 40*, differences in the *Elec* group are comparably small. In contrast, the *H2* and *E-Fuel* scenarios show higher electricity demand than the *Elec* scenarios. This is due to the higher amounts of electricity required for the production of synthetic, electricity-based energy carriers and the related conversion losses as well as the lower efficiency of hydrogen and methane-based heating systems compared to heat pumps.

In all scenarios, the climate-neutral electricity system in 2050 is vastly dominated by on- and offshore wind and photovoltaics. The system is supplemented by nuclear energy, concentrated solar power (CSP), hydropower and biomass. The high proportion of fluctuating renewable electricity production increases the need for flexibility to balance the energy supply. Energy storage, international trading and the demand-side flexibility offered by heat pumps (decentralised and large-

scale heat pumps in district heating), electric mobility as well as hydrogen production provide the necessary flexibility in the scenarios.

In terms of capacities, between 2020 and 2050, wind onshore capacity has to increase up to 789 GW in the *E-Fuel 80* scenario.<sup>5</sup> For photovoltaics, the capacity increase needed between 2020 and 2050 is up to 1147 GW in the *H2 80* scenario. Offshore wind is similar in all scenarios, due to the assumed resource constraints in line with the EU Offshore Renewable Energy Strategy that foresees 300 GW in 2050 [47], which means an increase of 286 GW from 2020 levels.

Table 2 shows the demand for hydrogen and synthetic methane in the EU-27 in 2050. It also lists the share of the demand that requires imports from outside Europe. The total demand covers the use of hydrogen or synthetic methane in buildings, district heating, and electricity as well as other sectors such as transport and industry.

<sup>5</sup> Capacities in 2020 are based on Eurostat data [NRG\_INF\_EPCRW].

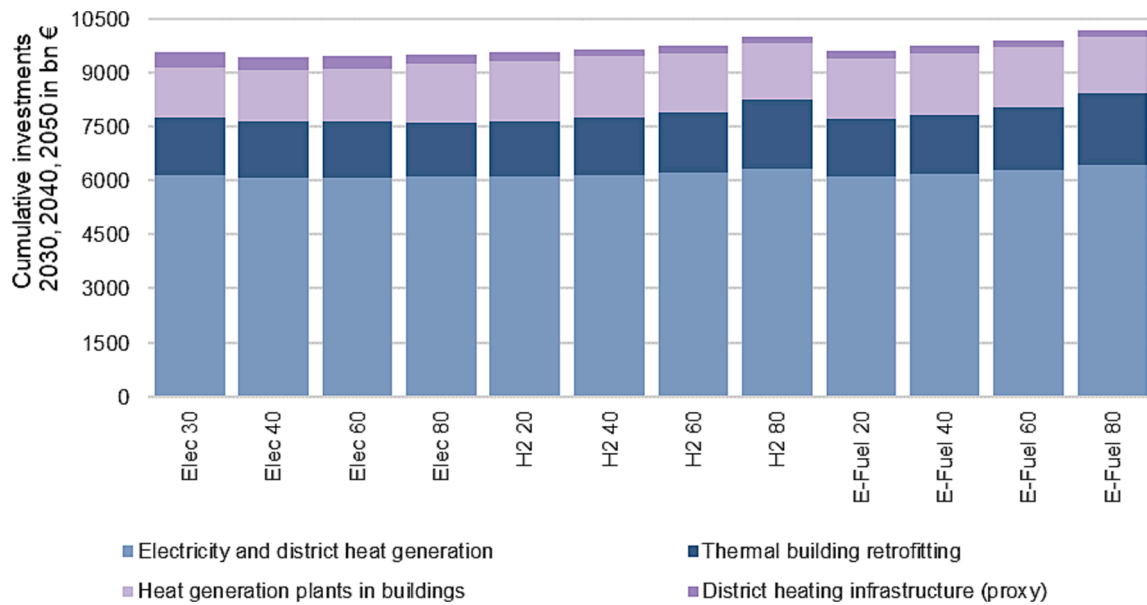


Fig. 9. Cumulative investment requirements in EU-27.

ENERTILE models the production of hydrogen and synthetic methane in Europe to cover demand and their generation costs. Alternatively, both energy carriers can be imported from outside of Europe (i.e. from the MENA region) at fixed costs (compare e.g. [36,48]).<sup>6</sup> The modelling results show that almost no import of hydrogen takes place because hydrogen can be produced cost-efficiently within Europe. In contrast, around 80% of the synthetic methane demand in 2050 across all scenarios is produced in EU-27, while the remaining 20% is imported from outside of Europe. Table 2 shows that the *E-Fuel* scenarios, especially the *E-Fuels 60* and *80* scenarios, imply a higher import dependence.

### 3.4. System costs and investment needs

Fig. 7 shows the annualised system costs in EU-27 under the different scenarios. The system costs displayed include capital costs as well as fixed and variable operation and maintenance (O&M) costs from both models. INVERT/OPT provides costs of refurbishment and decentralised heat technologies. ENERTILE provides the costs of conventional and renewable power plants, CHP plants, conventional and renewable heating technologies in district heating networks, cross-border electricity transmission capacities, energy storage technologies, hydrogen generation technologies and infrastructures, and e-fuel generation technologies. An estimation is included for district heating infrastructure costs, as the expansion of district heating varies substantially between scenarios. This estimation was calculated with specific costs obtained using the Hotmaps district heating expansion module<sup>7</sup>. In Fig. 7, the average of the system costs (per decade) in the scenarios serves as a baseline for the comparison. Thus, only the additional costs are displayed to make the differences between the scenarios more obvious. For example, in 2030, the cost in the *E-Fuel 80* scenario is €10 bn more than the average costs of all scenarios.

Fig. 7 shows that the cost differences between the scenarios are quite low in the period up until 2030, but increase substantially in later years (around €130 bn (23%) difference between *Elec 60* and *E-Fuel 80*

scenarios and around €70 bn (11%) between *Elec 60* and *H2 80* scenarios in 2050). In 2050, the overall lowest system costs are achieved in the *Elec 60* scenario. Hence, the cost-optimal scenario for the electrification of space and water heating is based on high shares of direct electric heating. In general, scenarios with lower usage of hydrogen or e-fuels in the heating sector have lower costs in 2050 compared to the average.

Furthermore, Fig. 8 details the cost components of the system costs in 2050 compared to the average costs of all scenarios. Variable costs of decentralised heating have the highest impact on overall costs, especially due to the high costs of hydrogen or e-fuels. Capital costs for decentralised heating are also important. O&M costs for electricity and district heating have comparatively little impact, while capital costs for the supply stream are relevant, especially in scenarios with a high penetration of hydrogen or e-fuels. District heating infrastructure costs have the highest impact in the *Elec 30* scenario, which is explained by the fact that this scenario has by far the largest district heating expansion.

Finally, Fig. 9 shows the cumulative investment requirements for all scenarios within the EU-27. The differences between the scenarios are moderate (around 6% between *Elec 60* and *H2 80*). Nevertheless, all scenarios with low use of hydrogen or e-fuels involve lower investment requirements. The investments are highest in the *H2 80* and *E-Fuel 80* scenarios, which is mainly due to the higher electricity generation required for the production of hydrogen or e-fuels. The cumulative investments are lowest in the *Elec 40* scenario, closely followed by the *Elec 60* scenario.

## 4. Discussion

The results show that there are some commonalities, but also clear differences between the scenarios. Developments that are robust across all scenarios are a strong uptake in renewable electricity generation and a substantial uptake of building renovations and thus decreased energy demand. These results are in line with other scenario-based modelling analyses (compare, e.g. meta-analyses in [4–6]). Thus, the expansion of renewable electricity generation and building retrofitting are key elements of the transition towards a climate-neutral space heating sector and, more broadly, towards a climate-neutral integrated energy system by 2050.

Regarding the differences, the analysis shows that the direct electrification of space and water heating, mainly via heat pumps, leads to 11% lower total system costs in 2050 as well as 6% lower (cumulative)

<sup>6</sup> Assumed hydrogen and synthetic methane import cost: Hydrogen for 100 €/MWh in 2030, 90 €/MWh in 2040 and 80 €/MWh in 2050; Synthetic methane for 160 €/MWh in 2030, 145 €/MWh in 2040 and 130 €/MWh in 2050.

<sup>7</sup> See [49] and <https://www.hotmaps-project.eu/>, <https://www.hotmaps-hevs.ch/>, <https://gitlab.com/hotmaps/heat>.

investments compared to indirect electrification with hydrogen. In the analysis, the direct electrification scenarios are the most cost-efficient pathways. The lowest costs are reached in the *Elec 60* scenario. The *Elec 80* scenario shows slightly higher costs than the *Elec 60* scenario because heat pumps are also pushed into areas with high heat densities, i.e. areas where district heating would be cost-efficient. In contrast, the *Elec 40* scenario shows higher costs than the *Elec 60* scenario due to lower availability of heat pumps and thus higher shares of biomass and hydrogen.

The results of our analysis also indicate that heat pumps are cost-efficient in decentralised systems and centralised district heating systems (in line with e.g. [8,17,25]). The ideal distribution between direct decentralised electric heating using heat pumps and direct centralised electric heating using large-scale heat pumps in district heating is therefore in the range of the *Elec 60* scenario.

Moreover, in all scenarios, renewable heat sources such as solar thermal and geothermal energy increase their share compared to today's levels (compare current levels in [46]). As our assumptions limit the use of these technologies to certain country-specific shares, higher shares of renewable heat might be possible and may even lead to lower overall system costs, which is partly analysed in [8] and should be focused on in further research.

Furthermore, in the scenario with the lowest costs, there is a limited need for hydrogen in 2050 (in line with [25]). In district heating, hydrogen-based heating technologies achieve relatively high capacities, even though their generation is low. This indicates that hydrogen plays a role as a backup technology for district heating. In decentralised heating systems, hydrogen and e-fuels have an almost negligible role. In the cost-efficient direct electrification scenarios, hydrogen and e-fuels in decentral systems cover (together) less than 1.5% of the energy needs in buildings in 2050.

In addition, the results show that the cost-efficient direct electrification scenarios have slightly lower building renovation requirements compared to the *H2* or *E-Fuel* scenarios. This may seem counterintuitive, as it is often stated that hydrogen and e-fuels can be used in buildings that are difficult to retrofit, while heat pumps are mainly suitable for modern or retrofitted buildings (compare e.g. [50,51]). However, hydrogen and e-fuels have the highest variable energy prices (in line with [45]), which in turn trigger renovation activities. A more detailed analysis of different renovation scenarios was not part of this paper but would be highly relevant for further research (building on e.g. [27,52,53]).

Scenarios with higher shares of hydrogen and e-fuels also lead to either higher renewable electricity generation or higher import requirements or a combination of both. Thus, it can be concluded that directly electrifying a substantial amount of the heating demand of buildings via heat pumps is not only beneficial in terms of costs but also with regards to renewable electricity generation, related infrastructure, import dependency and energy security.

In light of the current crisis and the EU's efforts to reduce gas consumption and thus its dependence on Russia, scenarios with low import requirements become even more appealing. As the modelling work in this paper was done before the start of the current energy crisis, the assumed fossil fuel prices do not reflect recent price increases. However, higher fossil fuel prices, in particular, higher natural gas prices would have only a limited impact on the results. This is mainly due to the long-term perspective of the scenarios and the phase-out of fossil fuels to reach climate neutrality. Higher prices would most likely lead to a faster phase-out, shifting transformation costs to earlier years. Analyses focusing on the impact of fuel price increases, especially for the year 2030, should be a focus of future research.

The results described and discussed in this paper refer to the EU-27

level. At member state level, the overall picture is similar with very few exceptions. In 2050, the direct electrification scenarios have the lowest costs in almost all member states. Due to country-specific characteristics, minor deviations are possible, but these do not affect the general findings of the analysis. In the scenarios, the target share is implemented at member state level, thus, all member states have to reach the same share in 2050 (compare section 2.1). A variation of shares could lead to lower costs, however, no major differences are to be expected. Selected member state results are presented in the appendix (compare Figure A 3 - A 5). Nonetheless, country-specific and regional analyses are highly relevant for further research (and could build on country-specific analyses, e.g. [36,54,55]). These analyses should examine local conditions in detail and look more closely at national renewable heat potentials as well as the potential deployment of hydrogen.

Lastly, it is important to highlight that several barriers and practical challenges need to be overcome to reach high levels of direct electrification in line with the scenario with the lowest costs (*Elec 60*). Renewable electricity plants as well as electric heating technologies are still associated with a variety of barriers. For example, for heat pumps, barriers range from high investment and operating costs to permitting delays and a lack of regulation and standardisation as well as a lack of knowledge of installers or a biased perception of consumers (compare e.g. [56–60]). Overcoming the barriers requires a holistic policy mix that includes economic, regulatory and information-based measures, which should be a focus of further research.

## 5. Conclusions

This paper presents alternative decarbonisation pathways for the electrification of space and water heating by 2050. It uses an innovative modelling approach that combines two highly detailed models enabling an in-depth quantitative analysis for the whole EU. The results contribute to the ongoing discussion on favourable pathways for heating buildings and the role of secondary energy carriers such as hydrogen. By doing this, the paper not only addresses a highly relevant research issue that is understudied today but also develops potentially highly relevant insights for policymaking.

Based on the comparison of the 12 scenarios, it can be concluded that direct electrification of space and hot water heating, mainly via heat pumps, leads to lower overall system costs compared to indirect electrification via hydrogen or e-fuels. Decarbonisation pathways based on large shares of heat pumps have 11% lower total system costs in 2050 than pathways based on larger shares of hydrogen and 23% lower total system costs than scenarios based on large shares of synthetic fuels. This translates into system cost savings of around €70 bn and €130 bn respectively.

The scenarios with large shares of direct electrification show slightly lower investments in building renovation, lower renewable electricity generation and lower levels of imports compared to scenarios with large shares of hydrogen or e-fuels. Hence, directly electrifying a substantial amount of the heating demand seems to be beneficial in terms of system costs but also with regard to thermal retrofitting investments, electricity generation and imports. The main research question can therefore be answered to the effect that, under consideration of the upstream supply sector, a large share of direct electrification is the most favourable electrification pathway to achieve decarbonised space and hot water heating in 2050.

To reach this favourable electrification pathway, a strong uptake of heat pumps is required. This applies to both decentralised heat pumps as well as centralised large-scale heat pumps in district heating systems. At the same time, the cost-efficient direct electrification scenarios foresee a

substantial expansion of district heating. Further, a fully decarbonised power sector with a substantial contribution of renewable sources, mainly wind and solar, must be achieved. In addition, a major increase in building renovations is indispensable. Together, these are the key elements of the transition to reach a climate-neutral energy system in Europe in 2050.

With these results, the paper gives robust guidance to policymakers to develop policy measures. It outlines that the most cost-efficient decarbonisation strategy for the space heating sector is based on three pillars: (1) strong and rapid deployment of heat pumps, (2) substantial uptake of building retrofitting and (3) rapid expansion of renewable electricity generation.

Further research focusing on cost-optimal pathways to decarbonise the heating and cooling sector is needed to facilitate the transition to a fully carbon-neutral heating and cooling supply. In particular, the potential contribution of renewable heat sources should be the focus of future analyses.

#### CRedit authorship contribution statement

**Anna Billerbeck:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Data curation, Visualization, Writing – original draft. **Christoph Kiefer:** Methodology, Software, Writing – review & editing. **Jenny Winkler:** Data curation, Writing – review & editing. **Christiane Bernath:** Methodology, Software, Writing – review & editing. **Frank Sensfuß:** Methodology, Software, Supervision. **Lukas Kranzl:** Methodology, Software, Writing – review & editing.

**Andreas Müller:** Methodology, Software, Writing – review & editing. **Mario Ragwitz:** Supervision.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

The authors do not have permission to share data.

#### Acknowledgements

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#### Appendix

(See Figs. A1–A5).

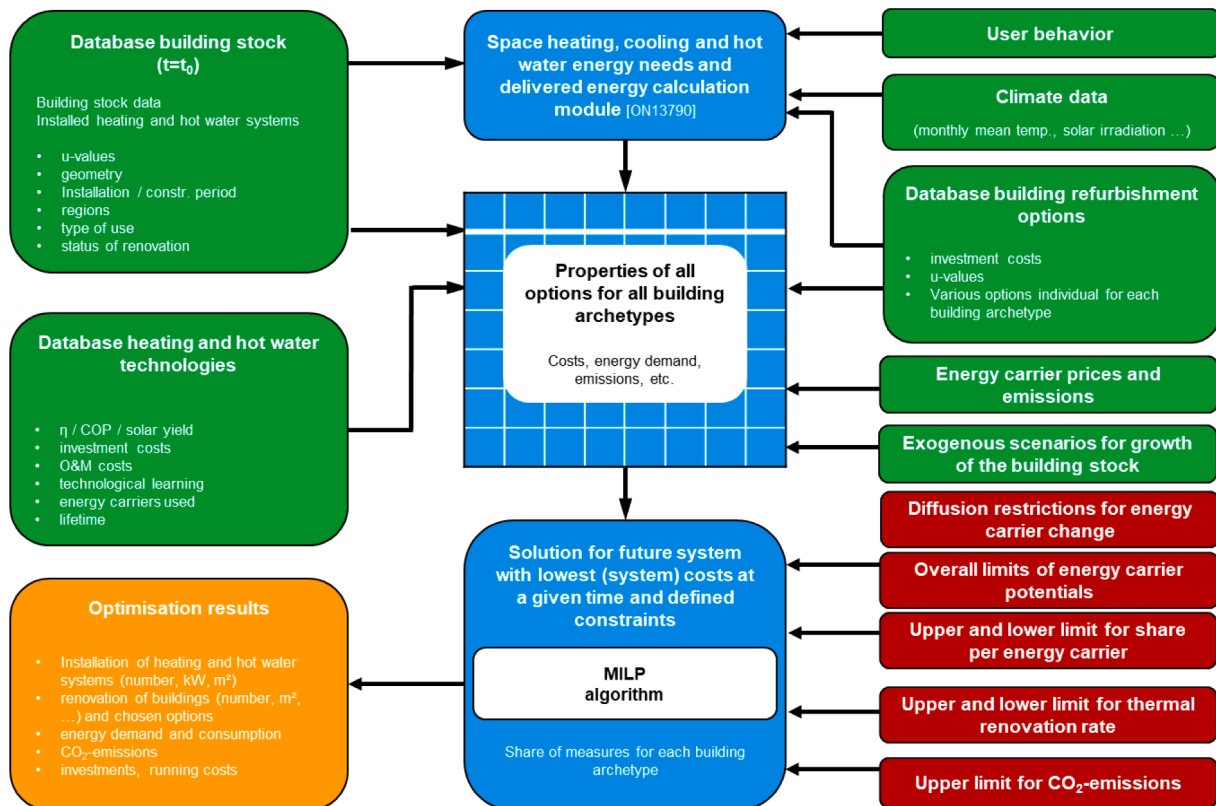


Fig. A1. Overview of the building model Invert/Opt (based on [27]).

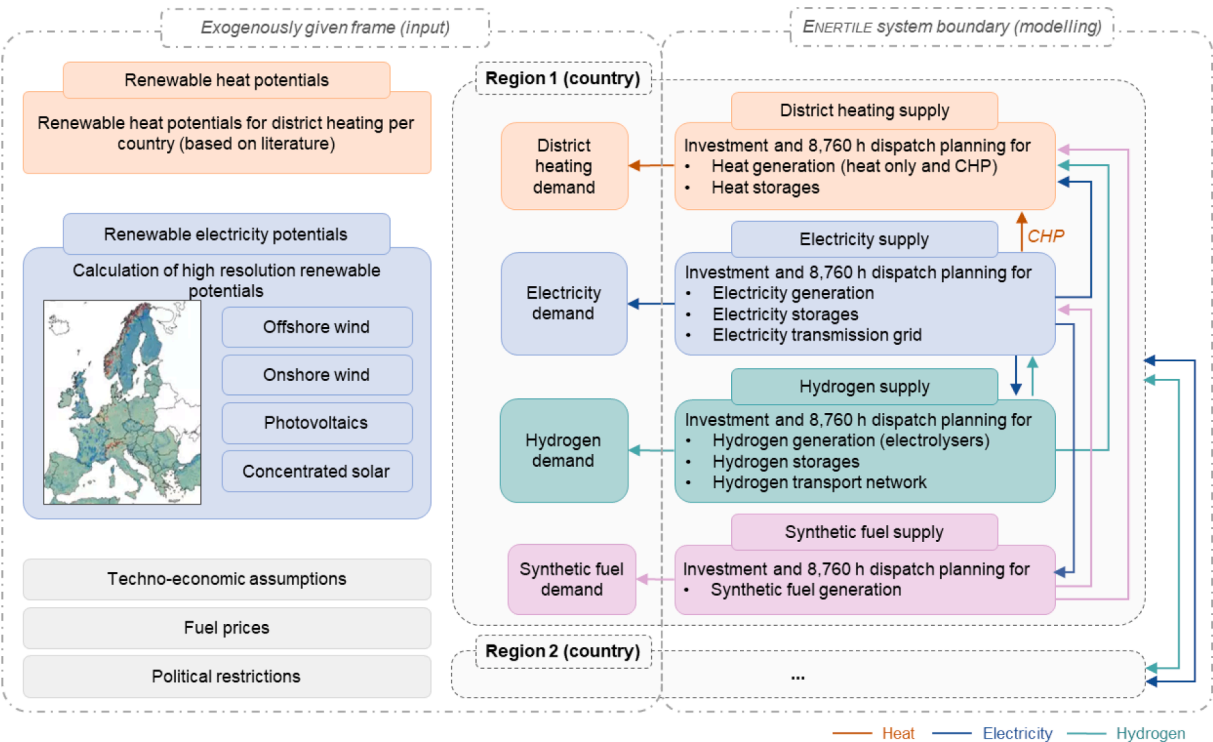


Fig. A2. Overview of the energy system model ENERTILE.

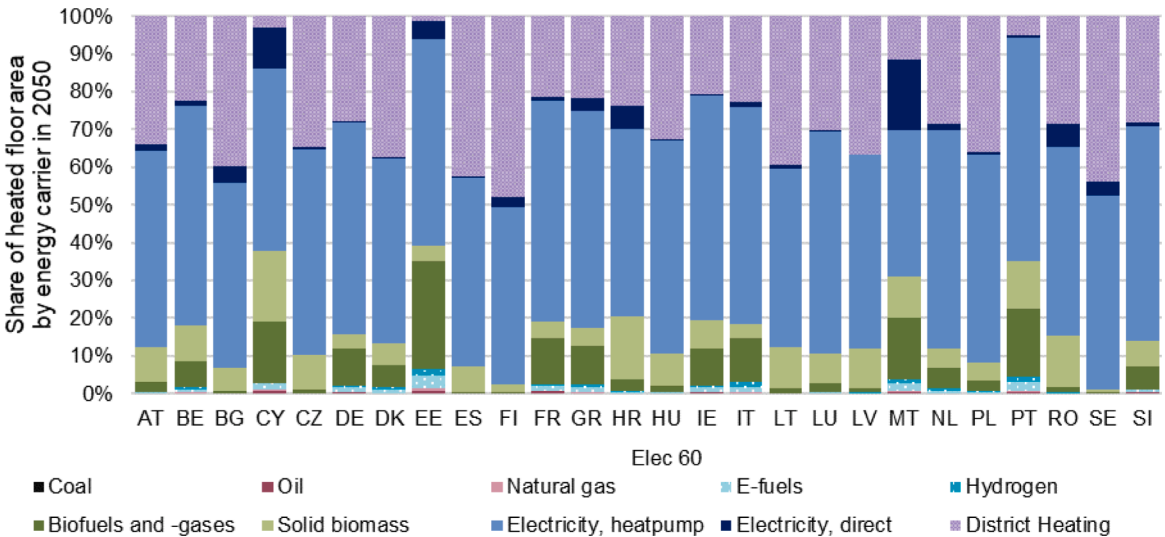


Fig. A3. Share of heated floor area by energy carrier in the member states in 2050 in Elec 60.

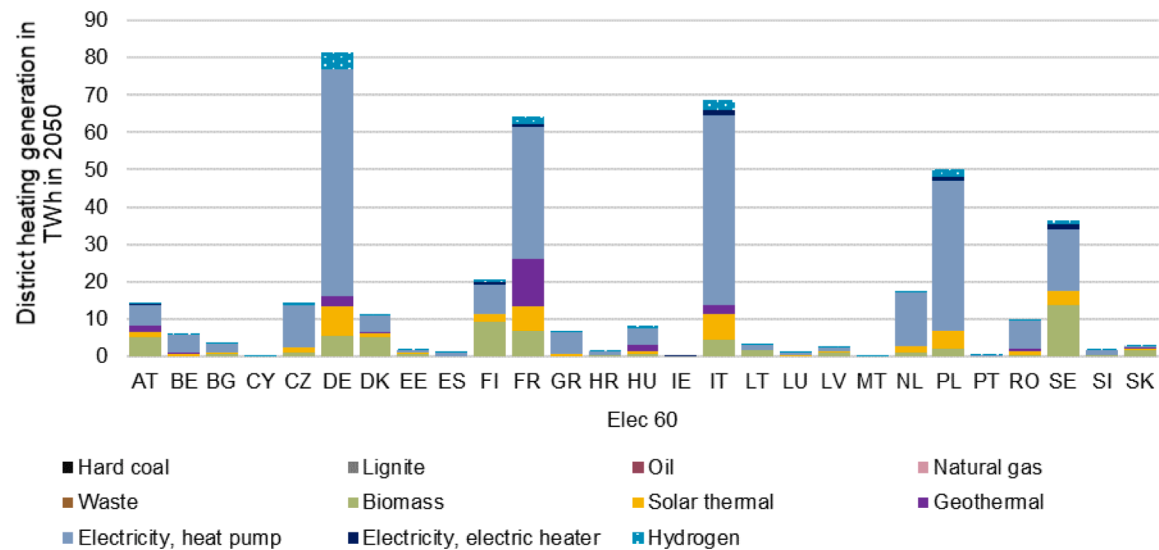


Fig. A4. District heating generation mix in the member states in 2050 in Elec 60.

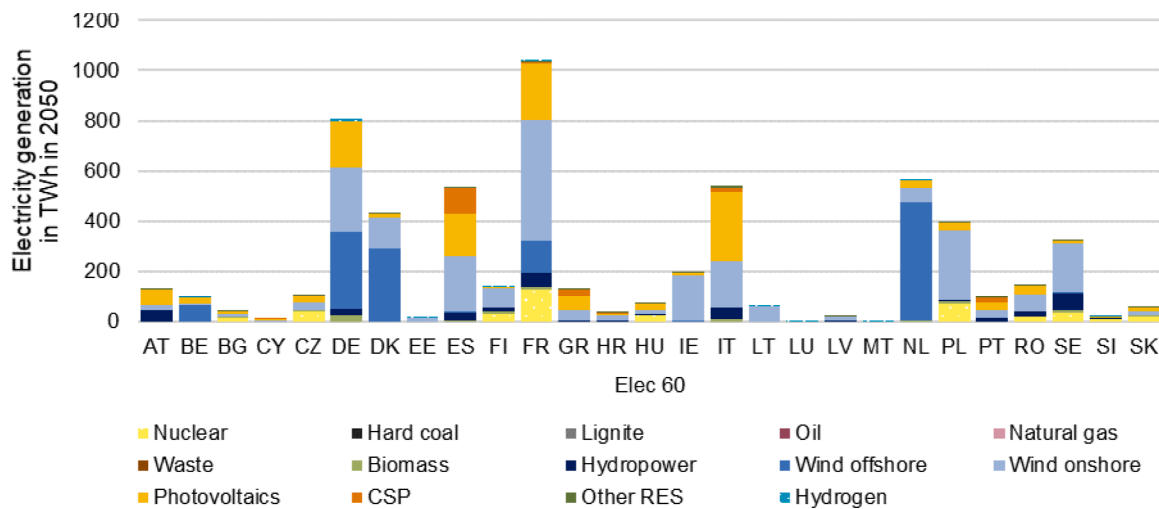


Fig. A5. Electricity generation mix in the member states in 2050 in Elec 60.

(See Table A1-A9).

**Table A1**  
Scenario settings and main assumptions.

| Parameter        |                                             | Scenario                                                                                                        |           |           |
|------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------|-----------|
|                  |                                             | Elec                                                                                                            | H2        | E-Fuel    |
| Buildings        | Thermal efficiency                          | upper and lower limit of buildings being retrofitted (assumption)                                               |           |           |
|                  | Biomass potentials                          | no growth of solid biomass resource use, country-specific upper limit of heated floor area (assumption)         |           |           |
|                  | Solar thermal                               | country-specific upper limit of heated floor area (assumption)                                                  |           |           |
|                  | Direct electric heating                     | 30%-80%                                                                                                         | optimised | optimised |
|                  | Heat pumps                                  | 30%-80%                                                                                                         | optimised | optimised |
|                  | Hydrogen                                    | optimised                                                                                                       | 20%-80%   | optimised |
|                  | E-fuels                                     | optimised                                                                                                       | optimised | 20%-80%   |
| District heating | Expansion of grid                           | country-specific upper limit of heated floor area derived from analysis of economic district heating potentials |           |           |
|                  | Biomass                                     | reduced deployment (assumption)                                                                                 |           |           |
|                  | Solar thermal                               | increasing deployment (assumption)                                                                              |           |           |
|                  | Geothermal                                  | increasing deployment (assumption)                                                                              |           |           |
|                  | Heat pumps                                  | optimised                                                                                                       |           |           |
|                  | Direct electric heating                     | optimised                                                                                                       |           |           |
|                  | Hydrogen                                    | optimised                                                                                                       |           |           |
| Electricity      | E-fuels                                     | optimised                                                                                                       |           |           |
|                  | Electricity demand in other sectors         | exogenously determined (assumption)                                                                             |           |           |
|                  | Hydrogen and e-fuel demand in other sectors | exogenously determined (assumption)                                                                             |           |           |
|                  | Biomass and waste                           | reduced deployment (assumption)                                                                                 |           |           |
|                  | Hydro and geothermal                        | current deployment (assumption)                                                                                 |           |           |
|                  | Other direct RES                            | optimised                                                                                                       |           |           |

**Table A2**  
Assumed fuel prices [61].

| Fuel        | Year | Price in €/MWh |
|-------------|------|----------------|
| Natural gas | 2030 | 22.23          |
|             | 2040 | 28.60          |
|             | 2050 | 31.44          |
| Hard coal   | 2030 | 9.59           |
|             | 2040 | 11.04          |
|             | 2050 | 12.13          |
| Oil         | 2030 | 44.32          |
|             | 2040 | 53.91          |
|             | 2050 | 65.29          |

**Table A3**  
Assumptions on investment costs for heating systems as seen by building owners for Germany (excluding subsidies). For heating systems, it is assumed that some share of the upfront investment costs related to connecting buildings are paid for by the utility and is then refinanced by energy consumption-independent annual fees (price basis: €2017) used in INVERT/OPT.

| Nominal capacity      | Investment costs (as seen by building ownersexcluding subsidies) [€/kW] |      |      |      |      |      |     |     |     |      |
|-----------------------|-------------------------------------------------------------------------|------|------|------|------|------|-----|-----|-----|------|
|                       | 7.5                                                                     | 10   | 12.5 | 15   | 25   | 50   | 100 | 200 | 500 | 1000 |
| District heating      | 261                                                                     | 196  | 180  | 169  | 142  | 106  | 80  | 61  | 50  | 46   |
| Oil condensing boiler | 736                                                                     | 552  | 514  | 489  | 392  | 265  | 205 | 142 | 105 | 91   |
| Gas condensing boiler | 431                                                                     | 323  | 294  | 274  | 226  | 164  | 123 | 94  | 77  | 71   |
| Wood log boiler       | 1142                                                                    | 856  | 788  | 743  | 623  | 474  |     |     |     |      |
| Wood chips boiler     | 2003                                                                    | 1502 | 1311 | 1183 | 911  | 676  | 550 | 425 | 350 | 323  |
| Wood pellets boiler   | 1463                                                                    | 1097 | 1040 | 1002 | 832  | 604  | 501 |     |     |      |
| Air/water heat pump   | 1205                                                                    | 904  | 867  | 843  | 773  | 660  | 581 | 479 |     |      |
| Brine/water heat pump | 1983                                                                    | 1487 | 1428 | 1389 | 1247 | 1040 | 923 | 728 | 612 | 570  |

**Table A4**

Assumptions on investment costs for refurbishment activities as seen by building owners for Germany (excluding subsidies; price basis: €2017) used in INVERT/OPT.

| Cost type                                | Base costs                        | Additional costs per cm insulation thickness                  | Cost of maintenance option        |
|------------------------------------------|-----------------------------------|---------------------------------------------------------------|-----------------------------------|
|                                          | [€/m <sup>2</sup> component area] | [€/cm insulation thickness and m <sup>2</sup> component area] | [€/m <sup>2</sup> component area] |
| <b>Buildings constructed since 1945</b>  |                                   |                                                               |                                   |
| Ceiling incl. roof                       | 190                               | 2.97                                                          | 136                               |
| Ceiling flat roof                        | 174                               | 0.70                                                          | 118                               |
| Exterior walls                           | 96                                | 2.67                                                          | 37                                |
| Building base                            | 29                                | 1.14                                                          | 0                                 |
| <b>Buildings constructed before 1945</b> |                                   |                                                               |                                   |
| Exterior walls                           | 125                               | 3.47                                                          | 48                                |
| Building base                            | 38                                | 1.48                                                          | 0                                 |
| Ceiling                                  | 228                               | 3.56                                                          | 163                               |

**Table A5**

Assumptions on investment costs for refurbishment activities (windows) as seen by building owners for Germany (excluding subsidies; price basis: €2017) used in INVERT/OPT.

| U-Value [W/m <sup>2</sup> K]      | 2.7                               | 1.7 | 1.3 | 0.95 | 0.65  | Maintenance costs |
|-----------------------------------|-----------------------------------|-----|-----|------|-------|-------------------|
| Unit                              | [€/m <sup>2</sup> component area] |     |     |      |       |                   |
| Residential                       |                                   | 359 | 379 | 427  | 475   | 201               |
| Non-residential                   |                                   | 285 | 308 | 367  | 425   | 160               |
| Buildings constructed before 1945 | 461                               | 467 | 492 | 555  | 1,190 | 216               |

**Table A6**Construction cost index for transferring prices between countries (Note: Based on <https://constructioncosts.eu/cost-index/> 2020/2022.)

| Country        | Construction Cost Index factor compared to Germany |
|----------------|----------------------------------------------------|
| Denmark        | 150%                                               |
| Portugal       | 52%                                                |
| Austria        | 104%                                               |
| Romania        | 48%                                                |
| Czech Republic | 63%                                                |
| Italy          | 97%                                                |
| Germany        | 100%                                               |
| Belgium        | 92%                                                |
| Bulgaria       | 50%                                                |
| Croatia        | 57%                                                |
| Cyprus         | 62%                                                |
| Estonia        | 61%                                                |
| Finland        | 118%                                               |
| France         | 108%                                               |
| Greece         | 66%                                                |
| Hungary        | 55%                                                |
| Ireland        | 82%                                                |
| Latvia         | 60%                                                |
| Lithuania      | 61%                                                |
| Luxembourg     | 102%                                               |
| Malta          | 82%                                                |
| Netherlands    | 85%                                                |
| Poland         | 68%                                                |
| Slovakia       | 53%                                                |
| Slovenia       | 83%                                                |
| Spain          | 73%                                                |
| Sweden         | 139%                                               |

**Table A7**

Techno-economic parameters of investment options in ENERTILE [62,63] (Note: The capacity expansion of nuclear, geothermal and solar thermal plants was set exogenously for each modelled country, thus, no costs are given. Coal phase-out plans were considered at country level. Fossil fuel technologies were given reduced lifetimes in order to achieve climate neutrality in 2050.)

| Technology       |                                 | Year | Lifetime | Investment in €/kW | Fixed O&M in €/kW | Variable O&M in €/MWh | Efficiency <i>el (chp)</i> |
|------------------|---------------------------------|------|----------|--------------------|-------------------|-----------------------|----------------------------|
| Electricity      | Coal steam plant                | 2030 | 20       | 1600               | 25.6              | 2.4                   | 43%                        |
|                  |                                 | 2030 | 20       | 775                | 11.6              | 3.0                   | 60%                        |
|                  | Combined cycle gas turbine      | 2040 | 10       | 750                | 11.3              | 3.0                   | 60%                        |
|                  |                                 | 2030 | 20       | 400                | 7.5               | 1.5                   | 41%                        |
|                  | Gas turbine                     | 2040 | 10       | 400                | 7.5               | 1.5                   | 41%                        |
|                  |                                 | 2030 | 10       | 228                | 6.3               | 0.0                   | 95%                        |
|                  | Battery storage                 | 2040 | 10       | 216                | 5.9               | 0.0                   | 95%                        |
|                  |                                 | 2050 | 10       | 204                | 5.5               | 0.0                   | 95%                        |
|                  |                                 | 2030 | 30       | 775                | 11.6              | 3.0                   | 59%                        |
|                  | Combined cycle hydrogen turbine | 2040 | 30       | 750                | 11.3              | 3.0                   | 60%                        |
|                  |                                 | 2050 | 30       | 750                | 11.3              | 3.0                   | 61%                        |
|                  |                                 | 2030 | 30       | 400                | 7.5               | 1.5                   | 41%                        |
|                  | Hydrogen turbine                | 2040 | 30       | 400                | 7.5               | 1.5                   | 41%                        |
|                  |                                 | 2050 | 30       | 400                | 7.5               | 1.5                   | 41%                        |
| CHP              | Combined cycle gas turbine CHP  | 2030 | 20       | 950                | 30                | 3.5                   | 48% (88%)                  |
|                  |                                 | 2040 | 10       | 950                | 30                | 3.5                   | 48% (88%)                  |
|                  | Gas turbine CHP                 | 2030 | 20       | 730                | 30                | 2.7                   | 33% (85%)                  |
|                  |                                 | 2040 | 10       | 730                | 30                | 2.7                   | 33% (85%)                  |
|                  | Biomass CHP                     | 2030 | 25       | 3200               | 115               | 2.1                   | 30% (71%)                  |
|                  |                                 | 2040 | 25       | 3050               | 109               | 2.1                   | 30% (71%)                  |
|                  |                                 | 2050 | 25       | 2900               | 103               | 2.1                   | 30% (71%)                  |
|                  | Hydrogen CHP                    | 2030 | 30       | 950                | 30                | 3                     | 48% (88%)                  |
|                  |                                 | 2040 | 30       | 950                | 30                | 3                     | 48% (88%)                  |
|                  |                                 | 2050 | 30       | 950                | 30                | 3                     | 48% (88%)                  |
| District heating | Gas boiler                      | 2030 | 19       | 50                 | 1.9               | 0.9                   | 104%                       |
|                  |                                 | 2040 | 9        | 50                 | 1.8               | 0.9                   | 104%                       |
|                  | Biomass boiler                  | 2030 | 25       | 830                | 47.9              | 0.7                   | 103%                       |
|                  |                                 | 2040 | 25       | 790                | 45.4              | 0.7                   | 103%                       |
|                  |                                 | 2050 | 25       | 750                | 42.9              | 0.7                   | 103%                       |
|                  | Electric boiler                 | 2030 | 20       | 60                 | 1.0               | 0.5                   | 99%                        |
|                  |                                 | 2040 | 20       | 60                 | 1.0               | 0.5                   | 99%                        |
|                  |                                 | 2050 | 20       | 60                 | 0.9               | 0.4                   | 99%                        |
|                  | Large-scale heat pump           | 2030 | 25       | 590                | 2.0               | 1.7                   | variable                   |
|                  |                                 | 2040 | 25       | 560                | 2.0               | 1.7                   | variable                   |
|                  |                                 | 2050 | 25       | 530                | 2.0               | 1.6                   | variable                   |
|                  | Heat storage                    | 2030 | 20       | 22                 | 0.0               | 0.0                   | 99%                        |
|                  |                                 | 2040 | 20       | 22                 | 0.0               | 0.0                   | 99%                        |
|                  |                                 | 2050 | 20       | 22                 | 0.0               | 0.0                   | 99%                        |
|                  | Hydrogen boiler                 | 2030 | 25       | 50                 | 1.9               | 0.9                   | 104%                       |
|                  |                                 | 2040 | 25       | 50                 | 1.9               | 0.9                   | 104%                       |
|                  |                                 | 2050 | 25       | 50                 | 1.9               | 0.9                   | 104%                       |

**Table A8**

Development of electricity demand in other sectors per country used in ENERTILE [64,65] (Note: The data publicly available from the 1.5TECH is limited to overall values for Europe in 2050 [64]. Thus, the development before 2050 and the demand distribution between countries were based on the SET-Nav project (diversification pathway) [65].)

| Country        | Demand in TWh     |      |       |                 |      |      |               |       |       |
|----------------|-------------------|------|-------|-----------------|------|------|---------------|-------|-------|
|                | Electric vehicles |      |       | Other transport |      |      | Other sectors |       |       |
|                | 2030              | 2040 | 2050  | 2030            | 2040 | 2050 | 2030          | 2040  | 2050  |
| Austria        | 0.7               | 3.3  | 7.7   | 3.1             | 3.2  | 3.3  | 65.6          | 73.0  | 78.9  |
| Belgium        | 1.0               | 4.3  | 10.1  | 1.9             | 2.2  | 2.4  | 80.5          | 94.0  | 109.3 |
| Bulgaria       | 0.1               | 0.9  | 2.8   | 0.7             | 0.6  | 0.6  | 29.6          | 31.7  | 34.3  |
| Switzerland    | 0.3               | 2.2  | 5.6   | 3.0             | 3.0  | 2.8  | 68.1          | 76.3  | 76.1  |
| Cyprus         | 0.0               | 0.1  | 0.3   | 0.0             | 0.0  | 0.0  | 3.3           | 3.9   | 4.7   |
| Czech Republic | 0.3               | 1.8  | 4.9   | 3.2             | 3.5  | 3.6  | 56.5          | 65.4  | 73.0  |
| Germany        | 5.1               | 35.7 | 101.4 | 23.2            | 39.2 | 59.9 | 511.7         | 542.6 | 573.1 |
| Denmark        | 1.2               | 3.5  | 7.3   | 0.8             | 2.0  | 3.8  | 33.7          | 38.1  | 42.1  |
| Estonia        | 0.1               | 0.3  | 0.9   | 0.1             | 0.1  | 0.2  | 7.3           | 7.8   | 8.3   |
| Spain          | 0.9               | 6.4  | 17.6  | 4.1             | 4.8  | 5.6  | 247.8         | 269.6 | 285.3 |
| Finland        | 0.4               | 2.1  | 5.6   | 1.4             | 1.9  | 2.7  | 76.3          | 84.5  | 89.8  |
| France         | 4.1               | 28.6 | 93.4  | 13.8            | 15.6 | 17.5 | 372.8         | 428.7 | 475.4 |
| Greece         | 0.4               | 2.4  | 5.9   | 0.1             | 0.1  | 0.1  | 44.9          | 50.1  | 53.1  |
| Croatia        | 0.1               | 0.4  | 1.1   | 0.5             | 0.5  | 0.6  | 13.3          | 14.4  | 15.3  |
| Hungary        | 0.1               | 1.0  | 2.7   | 1.8             | 1.9  | 1.9  | 39.0          | 44.2  | 47.7  |
| Ireland        | 0.2               | 1.0  | 2.7   | 0.1             | 0.1  | 0.2  | 23.3          | 26.7  | 29.9  |
| Italy          | 1.7               | 12.2 | 34.1  | 7.5             | 7.7  | 7.3  | 271.3         | 301.0 | 330.9 |
| Lithuania      | 0.1               | 0.3  | 0.9   | 0.5             | 0.5  | 0.5  | 10.2          | 11.1  | 12.5  |
| Luxembourg     | 0.0               | 0.2  | 0.5   | 0.1             | 0.1  | 0.1  | 7.6           | 8.9   | 9.2   |
| Latvia         | 0.0               | 0.2  | 0.6   | 0.3             | 0.4  | 0.5  | 7.5           | 8.5   | 8.9   |
| Malta          | 0.0               | 0.0  | 0.1   | –               | –    | –    | 2.1           | 2.3   | 2.4   |
| Netherlands    | 3.9               | 10.4 | 19.2  | 1.8             | 2.4  | 3.2  | 116.5         | 134.3 | 150.3 |
| Norway         | 3.0               | 7.7  | 14.4  | 0.9             | 1.1  | 1.5  | 132.4         | 148.3 | 148.0 |
| Poland         | 0.9               | 5.1  | 14.1  | 7.2             | 7.5  | 7.8  | 164.5         | 186.9 | 201.6 |
| Portugal       | 0.5               | 2.7  | 7.1   | 0.5             | 0.4  | 0.4  | 52.1          | 57.6  | 59.7  |
| Romania        | 0.3               | 1.7  | 5.6   | 2.5             | 2.8  | 2.9  | 52.5          | 61.2  | 68.9  |
| Sweden         | 3.2               | 12.3 | 22.7  | 3.2             | 4.0  | 5.3  | 120.1         | 132.1 | 141.9 |
| Slovenia       | 0.1               | 0.8  | 1.8   | 0.5             | 0.6  | 0.6  | 13.8          | 14.7  | 15.6  |
| Slovakia       | 0.1               | 0.5  | 1.8   | 1.4             | 1.4  | 1.4  | 29.6          | 34.7  | 36.2  |
| United Kingdom | 6.3               | 28.1 | 67.1  | 4.4             | 6.1  | 6.7  | 300.9         | 330.4 | 359.0 |
| Total          | 35                | 176  | 460   | 88              | 114  | 143  | 2,955         | 3,283 | 3,541 |

**Table A9**

Development of hydrogen, e-gas and e-liquids demand in other sectors per country used in ENERTILE [64,65] (Note: The data publicly available from the 1.5TECH is limited to overall values for Europe in 2050 [64]. Thus, the development before 2050 and the demand distribution between countries were based on the SET-Nav project (diversification pathway) [65].)

| Country        | Demand in TWh |      |       |       |      |      |           |      |      |
|----------------|---------------|------|-------|-------|------|------|-----------|------|------|
|                | Hydrogen      |      |       | E-gas |      |      | E-liquids |      |      |
|                | 2030          | 2040 | 2050  | 2030  | 2040 | 2050 | 2030      | 2040 | 2050 |
| Austria        | 1.3           | 9.2  | 27.0  | 0.5   | 3.4  | 10.2 | 0.9       | 6.1  | 18.0 |
| Belgium        | 3.9           | 26.7 | 74.5  | 1.5   | 10.1 | 28.1 | 2.6       | 17.9 | 49.9 |
| Bulgaria       | 0.5           | 3.1  | 7.8   | 0.2   | 1.2  | 3.0  | 0.3       | 2.1  | 5.2  |
| Switzerland    | 0.0           | 0.7  | 2.6   | 0.0   | 0.3  | 1.0  | 0.0       | 0.5  | 1.8  |
| Cyprus         | 0.0           | 0.1  | 0.2   | 0.0   | 0.0  | 0.1  | 0.0       | 0.0  | 0.1  |
| Czech Republic | 1.0           | 5.9  | 14.2  | 0.4   | 2.2  | 5.3  | 0.7       | 4.0  | 9.5  |
| Germany        | 9.8           | 57.2 | 143.6 | 3.7   | 21.6 | 54.1 | 6.6       | 38.3 | 96.1 |
| Denmark        | 0.0           | 0.1  | 0.3   | 0.0   | 0.0  | 0.1  | 0.0       | 0.0  | 0.2  |
| Estonia        | 0.0           | 0.2  | 0.5   | 0.0   | 0.1  | 0.2  | 0.0       | 0.1  | 0.4  |
| Spain          | 2.0           | 11.2 | 29.2  | 0.7   | 4.2  | 11.0 | 1.3       | 7.5  | 19.5 |
| Finland        | 0.5           | 2.7  | 6.6   | 0.2   | 1.0  | 2.5  | 0.3       | 1.8  | 4.4  |
| France         | 5.9           | 47.8 | 148.0 | 2.2   | 18.0 | 55.8 | 4.0       | 32.0 | 99.1 |
| Greece         | 0.2           | 2.0  | 6.5   | 0.1   | 0.7  | 2.4  | 0.1       | 1.3  | 4.3  |
| Croatia        | 0.1           | 0.4  | 0.8   | 0.0   | 0.1  | 0.3  | 0.1       | 0.2  | 0.5  |
| Hungary        | 0.9           | 4.8  | 11.8  | 0.3   | 1.8  | 4.5  | 0.6       | 3.2  | 7.9  |
| Ireland        | 0.1           | 0.8  | 2.4   | 0.0   | 0.3  | 0.9  | 0.0       | 0.5  | 1.6  |
| Italy          | 3.5           | 19.6 | 50.4  | 1.3   | 7.4  | 19.0 | 2.3       | 13.1 | 33.8 |
| Lithuania      | 0.4           | 2.2  | 5.5   | 0.1   | 0.8  | 2.1  | 0.3       | 1.5  | 3.7  |
| Luxembourg     | 0.0           | 0.0  | 0.0   | 0.0   | 0.0  | 0.0  | 0.0       | 0.0  | 0.0  |
| Latvia         | 0.0           | 0.0  | 0.0   | 0.0   | 0.0  | 0.0  | 0.0       | 0.0  | 0.0  |
| Malta          | 0.0           | 0.0  | 0.1   | 0.0   | 0.0  | 0.0  | 0.0       | 0.0  | 0.1  |
| Netherlands    | 4.0           | 22.6 | 56.7  | 1.5   | 8.5  | 21.3 | 2.7       | 15.1 | 37.9 |
| Norway         | 0.0           | 0.1  | 0.3   | 0.0   | 0.0  | 0.1  | 0.0       | 0.0  | 0.2  |
| Poland         | 1.9           | 9.8  | 23.4  | 0.7   | 3.7  | 8.8  | 1.3       | 6.5  | 15.7 |
| Portugal       | 0.5           | 3.0  | 7.4   | 0.2   | 1.1  | 2.8  | 0.4       | 2.0  | 5.0  |
| Romania        | 1.6           | 9.2  | 22.7  | 0.6   | 3.5  | 8.5  | 1.1       | 6.2  | 15.2 |
| Sweden         | 0.8           | 4.6  | 11.4  | 0.3   | 1.7  | 4.3  | 0.5       | 3.1  | 7.6  |
| Slovenia       | 0.0           | 0.0  | 0.0   | 0.0   | 0.0  | 0.0  | 0.0       | 0.0  | 0.0  |
| Slovakia       | 0.9           | 4.6  | 9.2   | 0.3   | 1.7  | 3.5  | 0.6       | 3.1  | 6.2  |
| United Kingdom | 3.2           | 17.6 | 44.1  | 1.2   | 6.6  | 16.6 | 2.1       | 11.8 | 29.5 |
| Total EU       | 43            | 266  | 707   | 16    | 100  | 266  | 29        | 178  | 473  |

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